

```
import sys
print(sys.executable)
C:\Users\pc\AppData\Local\Programs\Python\Python313\python.exe
import sys

!{sys.executable} -m pip install yfinance

Collecting yfinance
  Using cached yfinance-1.6.0-py3-none-any.whl.metadata (6.7 kB)
Requirement already satisfied: beautifulsoup4>=4.11.1 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from yfinance) (4.15.0)
Collecting curl_cffi>=0.15 (from yfinance)
  Using cached curl_cffi-0.16.0-cp310-abi3-win_amd64.whl.metadata (17 kB)
Collecting lxml>=4.9.0 (from yfinance)
  Using cached lxml-6.1.1-cp313-cp313-win_amd64.whl.metadata (3.6 kB)
Collecting multitasking>=0.0.7 (from yfinance)
  Using cached multitasking-0.0.13-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: numpy>=1.16.5 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from yfinance) (2.5.2)
Requirement already satisfied: pandas>=1.3.0 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from yfinance) (3.0.5)
Collecting peewee>=3.16.2 (from yfinance)
  Using cached peewee-4.3.0-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: platformdirs>=2.0.0 in c:\users\pc\appdata\roaming\python\python313\site-packages (from yfinance) (4.11.0)
Collecting protobuf>=3.19.0 (from yfinance)
  Downloading protobuf-7.35.1-cp310-abi3-win_amd64.whl.metadata (595 bytes)
Collecting pytz>=2022.5 (from yfinance)
  Downloading pytz-2026.3.post1-py2.py3-none-any.whl.metadata (22 kB)
Requirement already satisfied: requests>=2.31 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from yfinance) (2.34.2)
Requirement already satisfied: websockets>=13.0 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from yfinance) (15.0.1)
Requirement already satisfied: soupsieve>=1.6.1 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from beautifulsoup4>=4.11.1->yfinance) (2.9.1)
Requirement already satisfied: typing-extensions>=4.0.0 in c:\users\pc\appdata\roaming\python\python313\site-packages (from beautifulsoup4>=4.11.1->yfinance) (4.16.0)
```

Requirement already satisfied: cffi>=2.0.0 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from curl\_cffi>=0.15->yfinance) (2.1.0)  
Requirement already satisfied: certifi>=2024.2.2 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from curl\_cffi>=0.15->yfinance) (2026.7.22)  
Requirement already satisfied: pycparser in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from cffi>=2.0.0->curl\_cffi>=0.15->yfinance) (3.0)  
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\pc\appdata\roaming\python\python313\site-packages (from pandas>=1.3.0->yfinance) (2.9.0.post0)  
Requirement already satisfied: tzdata in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from pandas>=1.3.0->yfinance) (2026.3)  
Requirement already satisfied: six>=1.5 in c:\users\pc\appdata\roaming\python\python313\site-packages (from python-dateutil>=2.8.2->pandas>=1.3.0->yfinance) (1.17.0)  
Requirement already satisfied: charset\_normalizer<4,>=2 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.31->yfinance) (3.4.9)  
Requirement already satisfied: idna<4,>=2.5 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.31->yfinance) (3.18)  
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\pc\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.31->yfinance) (2.7.0)  
Using cached yfinance-1.6.0-py3-none-any.whl (148 kB)  
Using cached curl\_cffi-0.16.0-cp310-abi3-win\_amd64.whl (2.0 MB)  
Using cached lxml-6.1.1-cp313-cp313-win\_amd64.whl (4.0 MB)  
Using cached multitasking-0.0.13-py3-none-any.whl (16 kB)  
Using cached peewee-4.3.0-py3-none-any.whl (179 kB)  
Downloading protobuf-7.35.1-cp310-abi3-win\_amd64.whl (439 kB)  
Downloading pytz-2026.3.post1-py2.py3-none-any.whl (508 kB)  
Installing collected packages: pytz, multitasking, protobuf, peewee, lxml, curl\_cffi, yfinance

```
----- 0/7 [pytz]
----- 0/7 [pytz]
----- 0/7 [pytz]
----- 0/7 [pytz]
----- 0/7 [pytz]
----- 2/7 [protobuf]
----- 2/7 [protobuf]
----- 2/7 [protobuf]
----- 2/7 [protobuf]
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----- 2/7 [protobuf]
```

```

----- 2/7 [protobuf]
----- 3/7 [peewee]
----- 3/7 [peewee]
----- 3/7 [peewee]
----- 3/7 [peewee]
----- 3/7 [peewee]
----- 3/7 [peewee]
----- 4/7 [lxml]
----- 4/7 [lxml]
----- 4/7 [lxml]
----- 5/7 [curl_cffi]
----- 5/7 [curl_cffi]
----- 5/7 [curl_cffi]
----- 5/7 [curl_cffi]
----- 6/7 [yfinance]
----- 6/7 [yfinance]
----- 6/7 [yfinance]
----- 6/7 [yfinance]
----- 7/7 [yfinance]

```

Successfully installed curl\_cffi-0.16.0 lxml-6.1.1 multitasking-0.0.13  
peewee-4.3.0 protobuf-7.35.1 pytz-2026.3.post1 yfinance-1.6.0

WARNING: The script pwiz.exe is installed in 'C:\Users\pc\AppData\Local\Programs\Python\Python313\Scripts' which is not on PATH.  
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.  
WARNING: The script curl-cffi.exe is installed in 'C:\Users\pc\AppData\Local\Programs\Python\Python313\Scripts' which is not on PATH.  
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

[notice] A new release of pip is available: 25.1.1 -> 26.2.1  
[notice] To update, run: C:\Users\pc\AppData\Local\Programs\Python\Python313\python.exe -m pip install --upgrade pip

```

import yfinance as yf
import pandas as pd

print("yfinance is working!")

yfinance is working!

gold = yf.download(
    "GOLDBEES.NS",
    start="2009-01-01",
    end="2026-08-19",
    auto_adjust=False
)

gold.head()

```

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

| Price<br>\<br>Ticker | Adj Close<br>GOLDBEES.NS | Close<br>GOLDBEES.NS | High<br>GOLDBEES.NS | Low<br>GOLDBEES.NS | Open<br>GOLDBEES.NS |
|----------------------|--------------------------|----------------------|---------------------|--------------------|---------------------|
| Date                 |                          |                      |                     |                    |                     |
| 2009-01-02           | 13.3700                  | 13.3700              | 13.43               | 13.3000            | 0.0                 |
| 2009-01-05           | 13.1555                  | 13.1555              | 13.32               | 13.1500            | 0.0                 |
| 2009-01-06           | 13.0400                  | 13.0400              | 13.20               | 13.0201            | 0.0                 |
| 2009-01-07           | 13.2150                  | 13.2150              | 13.25               | 13.0600            | 0.0                 |
| 2009-01-09           | 13.1689                  | 13.1689              | 13.20               | 13.0600            | 0.0                 |

| Price<br>Ticker | Volume<br>GOLDBEES.NS |
|-----------------|-----------------------|
| Date            |                       |
| 2009-01-02      | 1043400               |
| 2009-01-05      | 1294700               |
| 2009-01-06      | 1236000               |
| 2009-01-07      | 1109000               |
| 2009-01-09      | 723900                |

```
print("Number of rows:", len(gold))
print("Start date:", gold.index.min())
print("End date:", gold.index.max())
print("\nColumns:")
print(gold.columns)
```

```
Number of rows: 4345
Start date: 2009-01-02 00:00:00
End date: 2026-08-18 00:00:00
```

```
Columns:
MultiIndex([('Adj Close', 'GOLDBEES.NS'),
            ('Close', 'GOLDBEES.NS'),
            ('High', 'GOLDBEES.NS'),
            ('Low', 'GOLDBEES.NS'),
            ('Open', 'GOLDBEES.NS'),
            ('Volume', 'GOLDBEES.NS')],
            names=['Price', 'Ticker'])
```

```
# Remove the second level of the column names
gold.columns = gold.columns.get_level_values(0)
```

```
# Check the columns
```

```
print(gold.columns)
```

```
Index(['Adj Close', 'Close', 'High', 'Low', 'Open', 'Volume'],  
      dtype='str', name='Price')
```

```
# Convert the Date index into a normal column
```

```
gold = gold.reset_index()
```

```
gold.head()
```

|   | Price | Date       | Adj Close | Close   | High  | Low     | Open | Volume  |
|---|-------|------------|-----------|---------|-------|---------|------|---------|
| 0 |       | 2009-01-02 | 13.3700   | 13.3700 | 13.43 | 13.3000 | 0.0  | 1043400 |
| 1 |       | 2009-01-05 | 13.1555   | 13.1555 | 13.32 | 13.1500 | 0.0  | 1294700 |
| 2 |       | 2009-01-06 | 13.0400   | 13.0400 | 13.20 | 13.0201 | 0.0  | 1236000 |
| 3 |       | 2009-01-07 | 13.2150   | 13.2150 | 13.25 | 13.0600 | 0.0  | 1109000 |
| 4 |       | 2009-01-09 | 13.1689   | 13.1689 | 13.20 | 13.0600 | 0.0  | 723900  |

```
gold.info()
```

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 4345 entries, 0 to 4344
```

```
Data columns (total 7 columns):
```

| # | Column    | Non-Null Count | Dtype         |
|---|-----------|----------------|---------------|
| 0 | Date      | 4345 non-null  | datetime64[s] |
| 1 | Adj Close | 4345 non-null  | float64       |
| 2 | Close     | 4345 non-null  | float64       |
| 3 | High      | 4345 non-null  | float64       |
| 4 | Low       | 4345 non-null  | float64       |
| 5 | Open      | 4345 non-null  | float64       |
| 6 | Volume    | 4345 non-null  | int64         |

```
dtypes: datetime64[s](1), float64(5), int64(1)
```

```
memory usage: 237.7 KB
```

```
gold.isnull().sum()
```

|           |       |
|-----------|-------|
| Price     |       |
| Date      | 0     |
| Adj Close | 0     |
| Close     | 0     |
| High      | 0     |
| Low       | 0     |
| Open      | 0     |
| Volume    | 0     |
| dtype:    | int64 |

```
print("Duplicate rows:", gold.duplicated().sum())
```

```
Duplicate rows: 0
```

```
silver = yf.download(
    "SILVERBEES.NS",
    start="2022-01-01",
    end="2026-08-19",
    auto_adjust=False
)
```

```
silver.head()
```

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

| Price<br>Ticker<br>Date | Adj Close<br>SILVERBEES.NS | Close<br>SILVERBEES.NS | High<br>SILVERBEES.NS | Low<br>SILVERBEES.NS |
|-------------------------|----------------------------|------------------------|-----------------------|----------------------|
| 2022-02-04              | 60.959599                  | 60.959599              | 60.959599             | 60.959599            |
| 2022-02-07              | 61.770000                  | 61.770000              | 64.339996             | 59.439999            |
| 2022-02-08              | 62.279999                  | 62.279999              | 63.000000             | 61.549999            |
| 2022-02-09              | 62.770000                  | 62.770000              | 63.000000             | 62.009998            |
| 2022-02-10              | 62.799999                  | 62.799999              | 64.900002             | 62.200001            |

| Price<br>Ticker<br>Date | Open<br>SILVERBEES.NS | Volume<br>SILVERBEES.NS |
|-------------------------|-----------------------|-------------------------|
| 2022-02-04              | 60.959599             | 0                       |
| 2022-02-07              | 60.000000             | 208834                  |
| 2022-02-08              | 62.000000             | 65666                   |
| 2022-02-09              | 62.500000             | 156227                  |
| 2022-02-10              | 63.000000             | 341078                  |

```
print("Number of rows:", len(silver))
print("Start date:", silver.index.min())
print("End date:", silver.index.max())
```

```
Number of rows: 1121
Start date: 2022-02-04 00:00:00
End date: 2026-08-18 00:00:00
```

```
silver.columns
```

```
MultiIndex([('Adj Close', 'SILVERBEES.NS'),
            ('Close', 'SILVERBEES.NS'),
            ('High', 'SILVERBEES.NS'),
            ('Low', 'SILVERBEES.NS'),
            ('Open', 'SILVERBEES.NS'),
            ('Volume', 'SILVERBEES.NS')],
           names=['Price', 'Ticker'])
```

```
silver.columns = silver.columns.get_level_values(0)
```

```
silver = silver.reset_index()
```

```
silver.head()
```

|   | Price<br>Open | Date       | Adj Close | Close     | High      | Low       |
|---|---------------|------------|-----------|-----------|-----------|-----------|
| 0 | 60.959599     | 2022-02-04 | 60.959599 | 60.959599 | 60.959599 | 60.959599 |
| 1 | 60.000000     | 2022-02-07 | 61.770000 | 61.770000 | 64.339996 | 59.439999 |
| 2 | 62.000000     | 2022-02-08 | 62.279999 | 62.279999 | 63.000000 | 61.549999 |
| 3 | 62.500000     | 2022-02-09 | 62.770000 | 62.770000 | 63.000000 | 62.009998 |
| 4 | 63.000000     | 2022-02-10 | 62.799999 | 62.799999 | 64.900002 | 62.200001 |

|   | Price | Volume |
|---|-------|--------|
| 0 |       | 0      |
| 1 |       | 208834 |
| 2 |       | 65666  |
| 3 |       | 156227 |
| 4 |       | 341078 |

```
# Remove the extra Ticker level from Gold columns
```

```
gold.columns = gold.columns.get_level_values(0)
```

```
# Convert Date from index to a normal column
```

```
gold = gold.reset_index()
```

```
# Make sure Date is datetime
```

```
gold["Date"] = pd.to_datetime(gold["Date"])
```

```
gold.head()
```

|   | Price   | index | Date       | Adj Close | Close   | High  | Low     | Open |
|---|---------|-------|------------|-----------|---------|-------|---------|------|
| 0 | 1043400 | 0     | 2009-01-02 | 13.3700   | 13.3700 | 13.43 | 13.3000 | 0.0  |
| 1 | 1294700 | 1     | 2009-01-05 | 13.1555   | 13.1555 | 13.32 | 13.1500 | 0.0  |
| 2 | 1236000 | 2     | 2009-01-06 | 13.0400   | 13.0400 | 13.20 | 13.0201 | 0.0  |
| 3 | 1109000 | 3     | 2009-01-07 | 13.2150   | 13.2150 | 13.25 | 13.0600 | 0.0  |
| 4 | 723900  | 4     | 2009-01-09 | 13.1689   | 13.1689 | 13.20 | 13.0600 | 0.0  |

```
# Remove the extra Ticker level from Silver columns
```

```
silver.columns = silver.columns.get_level_values(0)
```

```
# Convert Date from index to a normal column
```

```
silver = silver.reset_index()

# Make sure Date is datetime
silver["Date"] = pd.to_datetime(silver["Date"])
```

```
silver.head()
```

|   | Price | index | Date       | Adj Close | Close     | High      | Low       | \ |
|---|-------|-------|------------|-----------|-----------|-----------|-----------|---|
| 0 |       | 0     | 2022-02-04 | 60.959599 | 60.959599 | 60.959599 | 60.959599 |   |
| 1 |       | 1     | 2022-02-07 | 61.770000 | 61.770000 | 64.339996 | 59.439999 |   |
| 2 |       | 2     | 2022-02-08 | 62.279999 | 62.279999 | 63.000000 | 61.549999 |   |
| 3 |       | 3     | 2022-02-09 | 62.770000 | 62.770000 | 63.000000 | 62.009998 |   |
| 4 |       | 4     | 2022-02-10 | 62.799999 | 62.799999 | 64.900002 | 62.200001 |   |

|   | Price | Open      | Volume |
|---|-------|-----------|--------|
| 0 |       | 60.959599 | 0      |
| 1 |       | 60.000000 | 208834 |
| 2 |       | 62.000000 | 65666  |
| 3 |       | 62.500000 | 156227 |
| 4 |       | 63.000000 | 341078 |

```
print("GOLD")
print(gold.shape)
print(gold.dtypes)
```

```
print("\nSILVER")
print(silver.shape)
print(silver.dtypes)
```

```
GOLD
(4345, 8)
Price
index          int64
Date          datetime64[s]
Adj Close      float64
Close          float64
High           float64
Low            float64
Open           float64
Volume         int64
dtype: object
```

```
SILVER
(1121, 8)
Price
index          int64
Date          datetime64[s]
Adj Close      float64
Close          float64
High           float64
```

```
Low          float64
Open         float64
Volume       int64
dtype: object
```

```
print("Gold missing values:")
print(gold.isnull().sum())
```

```
print("\nSilver missing values:")
print(silver.isnull().sum())
```

Gold missing values:

```
Price
index      0
Date       0
Adj Close  0
Close      0
High       0
Low        0
Open       0
Volume     0
dtype: int64
```

Silver missing values:

```
Price
index      0
Date       0
Adj Close  0
Close      0
High       0
Low        0
Open       0
Volume     0
dtype: int64
```

```
print("Gold duplicates:", gold.duplicated().sum())
print("Silver duplicates:", silver.duplicated().sum())
```

```
Gold duplicates: 0
Silver duplicates: 0
```

*# Find the common date range*

```
common_start = max(gold["Date"].min(), silver["Date"].min())
common_end = min(gold["Date"].max(), silver["Date"].max())
```

```
print("Common Start Date:", common_start)
print("Common End Date:", common_end)
```

```
Common Start Date: 2022-02-04 00:00:00
Common End Date: 2026-08-18 00:00:00
```

```

gold_common = gold[
    (gold["Date"] >= common_start) &
    (gold["Date"] <= common_end)
].copy()

silver_common = silver[
    (silver["Date"] >= common_start) &
    (silver["Date"] <= common_end)
].copy()

print("Gold comparison rows:", len(gold_common))
print("Silver comparison rows:", len(silver_common))

Gold comparison rows: 1121
Silver comparison rows: 1121

print("Gold comparison rows:", len(gold_common))
print("Silver comparison rows:", len(silver_common))

Gold comparison rows: 1121
Silver comparison rows: 1121

gold_common["Daily_Return"] = (
    gold_common["Adj Close"].pct_change() * 100
)

gold_common[["Date", "Adj Close", "Daily_Return"]].head()

```

|      | Price      | Date      | Adj Close | Daily_Return |
|------|------------|-----------|-----------|--------------|
| 3224 | 2022-02-04 | 41.599998 | NaN       |              |
| 3225 | 2022-02-07 | 41.630001 | 0.072122  |              |
| 3226 | 2022-02-08 | 41.820000 | 0.456398  |              |
| 3227 | 2022-02-09 | 41.990002 | 0.406509  |              |
| 3228 | 2022-02-10 | 42.169998 | 0.428665  |              |

```

silver_common["Daily_Return"] = (
    silver_common["Adj Close"].pct_change() * 100
)

silver_common[["Date", "Adj Close", "Daily_Return"]].head()

```

|   | Price      | Date      | Adj Close | Daily_Return |
|---|------------|-----------|-----------|--------------|
| 0 | 2022-02-04 | 60.959599 | NaN       |              |
| 1 | 2022-02-07 | 61.770000 | 1.329408  |              |
| 2 | 2022-02-08 | 62.279999 | 0.825641  |              |
| 3 | 2022-02-09 | 62.770000 | 0.786772  |              |
| 4 | 2022-02-10 | 62.799999 | 0.047792  |              |

```

gold_common = gold_common.dropna(subset=["Daily_Return"]).copy()
silver_common = silver_common.dropna(subset=["Daily_Return"]).copy()

```

```

print("Gold rows:", len(gold_common))
print("Silver rows:", len(silver_common))

Gold rows: 1120
Silver rows: 1120

gold_common["Cumulative_Return"] = (
    (1 + gold_common["Daily_Return"] / 100).cumprod() - 1
) * 100

silver_common["Cumulative_Return"] = (
    (1 + silver_common["Daily_Return"] / 100).cumprod() - 1
) * 100

print(gold_common[["Date", "Cumulative_Return"]].tail())
print(silver_common[["Date", "Cumulative_Return"]].tail())

```

| Price | Date       | Cumulative_Return |
|-------|------------|-------------------|
| 4340  | 2026-08-11 | 201.730772        |
| 4341  | 2026-08-12 | 203.966350        |
| 4342  | 2026-08-13 | 201.634635        |
| 4343  | 2026-08-14 | 200.793274        |
| 4344  | 2026-08-18 | 203.918281        |

| Price | Date       | Cumulative_Return |
|-------|------------|-------------------|
| 1116  | 2026-08-11 | 264.700577        |
| 1117  | 2026-08-12 | 270.343637        |
| 1118  | 2026-08-13 | 262.485985        |
| 1119  | 2026-08-14 | 262.896093        |
| 1120  | 2026-08-18 | 263.552255        |

```

import matplotlib.pyplot as plt
import seaborn as sns

print("Visualization libraries loaded successfully!")

Visualization libraries loaded successfully!

plt.figure(figsize=(14, 6))

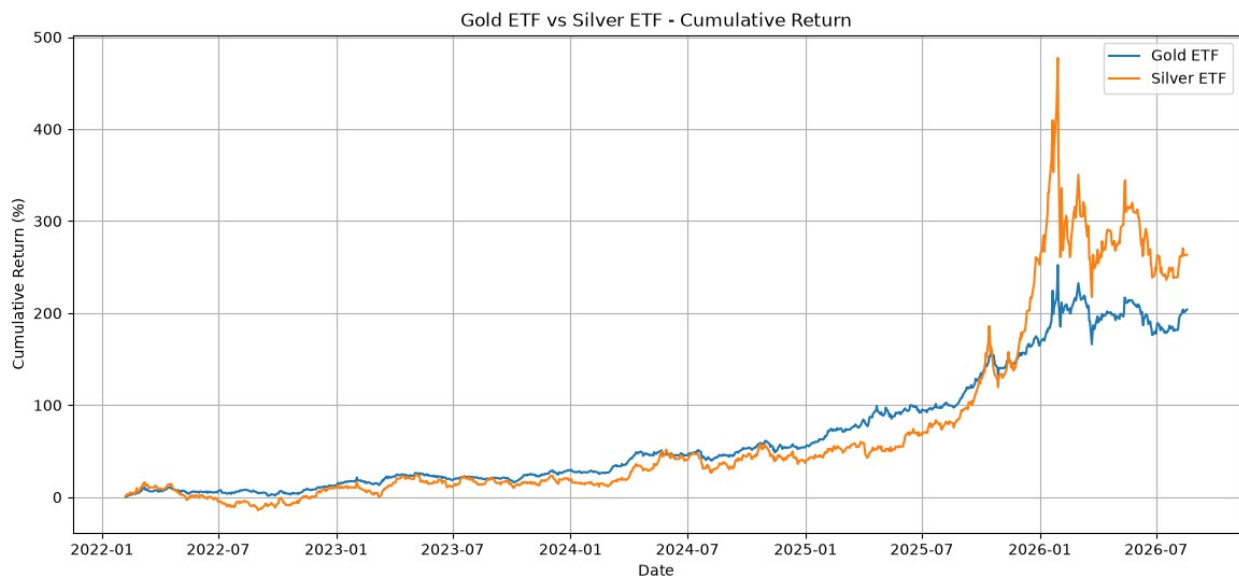
plt.plot(
    gold_common["Date"],
    gold_common["Cumulative_Return"],
    label="Gold ETF"
)

plt.plot(
    silver_common["Date"],
    silver_common["Cumulative_Return"],
    label="Silver ETF"
)

```

```
plt.title("Gold ETF vs Silver ETF - Cumulative Return")
plt.xlabel("Date")
plt.ylabel("Cumulative Return (%)")
plt.legend()
plt.grid(True)

plt.show()
```



```
gold_total_return = gold_common["Cumulative_Return"].iloc[-1]
silver_total_return = silver_common["Cumulative_Return"].iloc[-1]
```

```
print(f"Gold ETF Total Return: {gold_total_return:.2f}%")
print(f"Silver ETF Total Return: {silver_total_return:.2f}%")
```

```
Gold ETF Total Return: 203.92%
Silver ETF Total Return: 263.55%
```

```
initial_investment = 100000
```

```
gold_common["Investment_Value"] = (
    initial_investment *
    (1 + gold_common["Daily_Return"] / 100).cumprod()
)
```

```
silver_common["Investment_Value"] = (
    initial_investment *
    (1 + silver_common["Daily_Return"] / 100).cumprod()
)
```

```
print(
    f"Gold ₹1 Lakh Final Value: "
    f"₹{gold_common['Investment_Value'].iloc[-1]:,.2f}"
)
```

```

)
print(
    f"Silver ₹1 Lakh Final Value: "
    f"₹{silver_common['Investment_Value'].iloc[-1]:,.2f}"
)
Gold ₹1 Lakh Final Value: ₹303,918.28
Silver ₹1 Lakh Final Value: ₹363,552.25

gold_common["Year"] = gold_common["Date"].dt.year
silver_common["Year"] = silver_common["Date"].dt.year

gold_yearly = (
    gold_common
    .groupby("Year")["Daily_Return"]
    .apply(lambda x: ((1 + x / 100).prod() - 1) * 100)
)

silver_yearly = (
    silver_common
    .groupby("Year")["Daily_Return"]
    .apply(lambda x: ((1 + x / 100).prod() - 1) * 100)
)

import numpy as np

yearly_comparison = pd.DataFrame({
    "Gold_Return": gold_yearly,
    "Silver_Return": silver_yearly
})

yearly_comparison["Winner"] = np.where(
    yearly_comparison["Gold_Return"] >
    yearly_comparison["Silver_Return"],
    "Gold",
    "Silver"
)

yearly_comparison.round(2)

```

| Year | Gold_Return | Silver_Return | Winner |
|------|-------------|---------------|--------|
| 2022 | 12.48       | 10.29         | Gold   |
| 2023 | 14.68       | 7.36          | Gold   |
| 2024 | 19.44       | 16.00         | Gold   |
| 2025 | 71.81       | 157.29        | Silver |
| 2026 | 14.82       | 2.87          | Gold   |

```

yearly_comparison[["Gold_Return", "Silver_Return"]].plot(
    kind="bar",

```

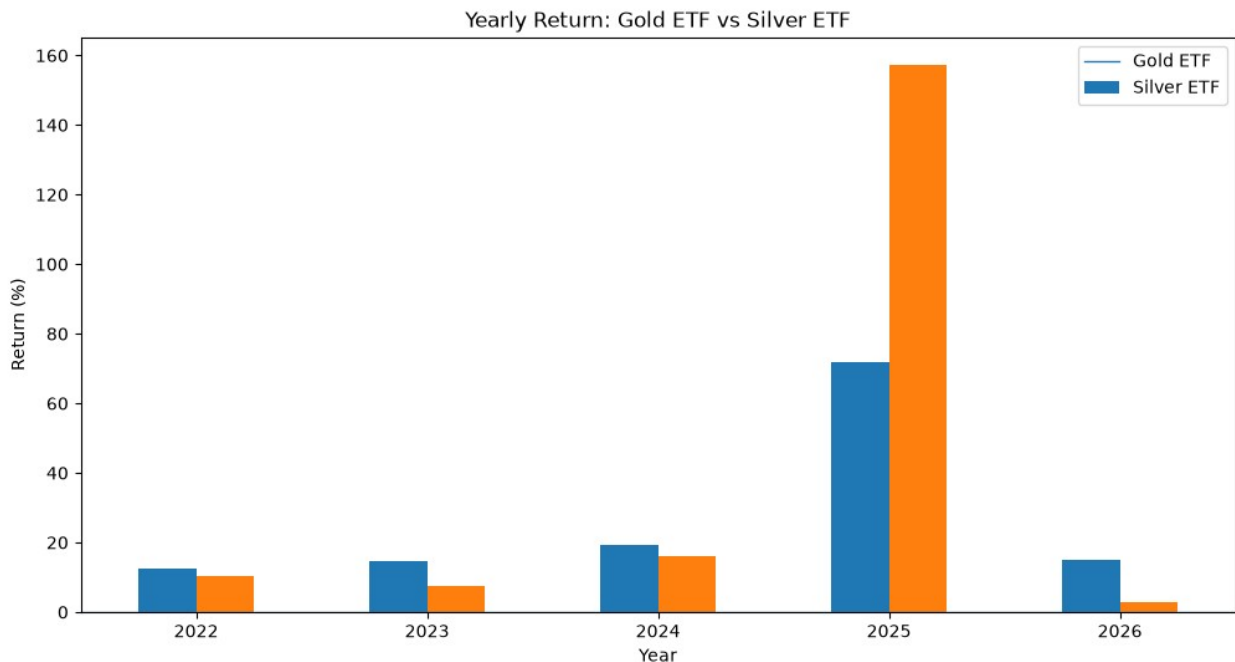
```

    figsize=(12, 6)
)

plt.title("Yearly Return: Gold ETF vs Silver ETF")
plt.xlabel("Year")
plt.ylabel("Return (%)")
plt.axhline(0, linewidth=1)
plt.legend(["Gold ETF", "Silver ETF"])
plt.xticks(rotation=0)

plt.show()

```



```

gold_volatility = gold_common["Daily_Return"].std()
silver_volatility = silver_common["Daily_Return"].std()

print(f"Gold Daily Volatility: {gold_volatility:.4f}%")
print(f"Silver Daily Volatility: {silver_volatility:.4f}%")

Gold Daily Volatility: 1.1912%
Silver Daily Volatility: 2.1635%

gold_annual_volatility = gold_volatility * np.sqrt(252)
silver_annual_volatility = silver_volatility * np.sqrt(252)

print(f"Gold Annualized Volatility: {gold_annual_volatility:.2f}%")
print(f"Silver Annualized Volatility: {silver_annual_volatility:.2f}%")

```

```

Gold Annualized Volatility: 18.91%
Silver Annualized Volatility: 34.34%

gold_common["Running_Peak"] = gold_common["Adj Close"].cummax()

gold_common["Drawdown"] = (
    (gold_common["Adj Close"] - gold_common["Running_Peak"])
    / gold_common["Running_Peak"]
) * 100

silver_common["Running_Peak"] = silver_common["Adj Close"].cummax()

silver_common["Drawdown"] = (
    (silver_common["Adj Close"] - silver_common["Running_Peak"])
    / silver_common["Running_Peak"]
) * 100

gold_max_drawdown = gold_common["Drawdown"].min()
silver_max_drawdown = silver_common["Drawdown"].min()

print(f"Gold Maximum Drawdown: {gold_max_drawdown:.2f}%")
print(f"Silver Maximum Drawdown: {silver_max_drawdown:.2f}%")

Gold Maximum Drawdown: -24.44%
Silver Maximum Drawdown: -45.01%

gold_volatility = gold_common["Daily_Return"].std()
silver_volatility = silver_common["Daily_Return"].std()

gold_annual_volatility = gold_volatility * np.sqrt(252)
silver_annual_volatility = silver_volatility * np.sqrt(252)

print(f"Gold Annualized Volatility: {gold_annual_volatility:.2f}%")
print(f"Silver Annualized Volatility: {silver_annual_volatility:.2f}%")

Gold Annualized Volatility: 18.91%
Silver Annualized Volatility: 34.34%

gold_common["Running_Peak"] = gold_common["Adj Close"].cummax()
gold_common["Drawdown"] = (
    (gold_common["Adj Close"] - gold_common["Running_Peak"])
    / gold_common["Running_Peak"]
) * 100

silver_common["Running_Peak"] = silver_common["Adj Close"].cummax()
silver_common["Drawdown"] = (
    (silver_common["Adj Close"] - silver_common["Running_Peak"])
    / silver_common["Running_Peak"]
) * 100

```

```
print(f"Gold Maximum Drawdown: {gold_common['Drawdown'].min():.2f}%")
print(f"Silver Maximum Drawdown: {silver_common['Drawdown'].min():.2f}%")
```

```
Gold Maximum Drawdown: -24.44%
Silver Maximum Drawdown: -45.01%
```

```
gold_volatility = gold_common["Daily_Return"].std()
silver_volatility = silver_common["Daily_Return"].std()
```

```
gold_annual_volatility = gold_volatility * np.sqrt(252)
silver_annual_volatility = silver_volatility * np.sqrt(252)
```

```
print(f"Gold Annualized Volatility: {gold_annual_volatility:.2f}%")
print(f"Silver Annualized Volatility: {silver_annual_volatility:.2f}%")
```

```
Gold Annualized Volatility: 18.91%
Silver Annualized Volatility: 34.34%
```

```
comparison = pd.merge(
    gold_common[["Date", "Daily_Return"]],
    silver_common[["Date", "Daily_Return"]],
    on="Date",
    suffixes=("_Gold", "_Silver")
)
```

```
comparison.head()
```

|   | Price | Date       | Daily_Return_Gold | Daily_Return_Silver |
|---|-------|------------|-------------------|---------------------|
| 0 |       | 2022-02-07 | 0.072122          | 1.329408            |
| 1 |       | 2022-02-08 | 0.456398          | 0.825641            |
| 2 |       | 2022-02-09 | 0.406509          | 0.786772            |
| 3 |       | 2022-02-10 | 0.428665          | 0.047792            |
| 4 |       | 2022-02-11 | 0.047428          | -0.700635           |

```
correlation = comparison[
    ["Daily_Return_Gold", "Daily_Return_Silver"]
].corr()
```

```
print(correlation)
```

|                     | Price | Daily_Return_Gold | Daily_Return_Silver |
|---------------------|-------|-------------------|---------------------|
| Price               |       |                   |                     |
| Daily_Return_Gold   |       | 1.00000           | 0.81316             |
| Daily_Return_Silver |       | 0.81316           | 1.00000             |

```
plt.figure(figsize=(6, 4))
```

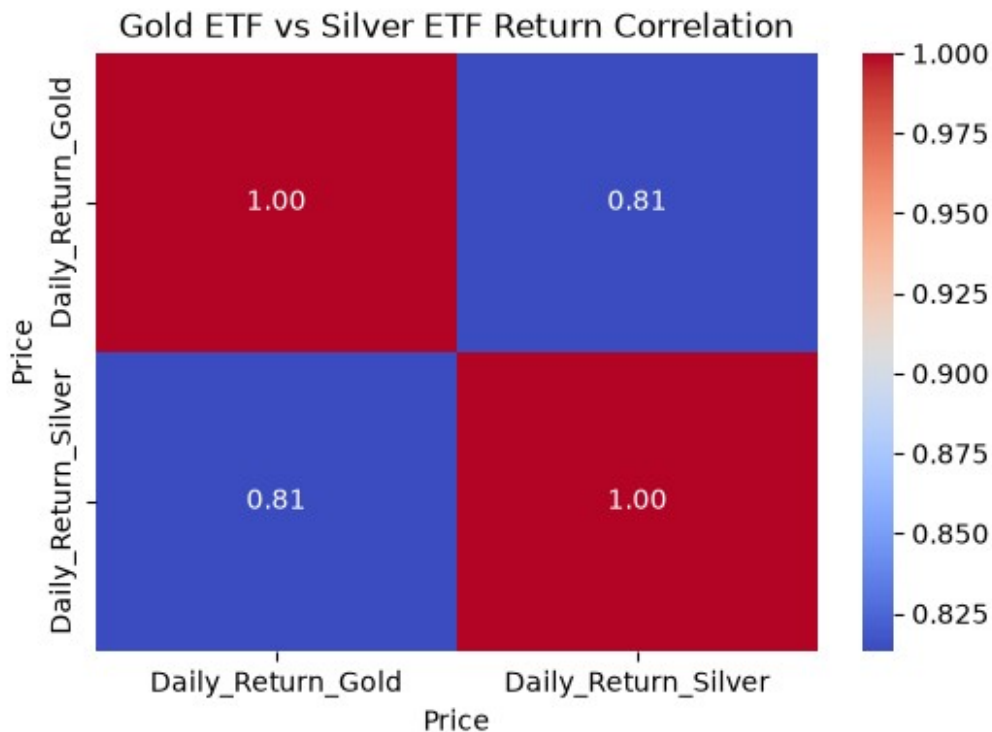
```
sns.heatmap(
    correlation,
    annot=True,
```

```

    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Gold ETF vs Silver ETF Return Correlation")
plt.show()

```



```

# Annualized volatility
gold_volatility = gold_common["Daily_Return"].std() * np.sqrt(252)
silver_volatility = silver_common["Daily_Return"].std() * np.sqrt(252)

print(f"Gold Annualized Volatility: {gold_volatility:.2f}%")
print(f"Silver Annualized Volatility: {silver_volatility:.2f}%")

Gold Annualized Volatility: 18.91%
Silver Annualized Volatility: 34.34%

# Correlation between Gold and Silver daily returns
comparison = pd.merge(
    gold_common[["Date", "Daily_Return"]],
    silver_common[["Date", "Daily_Return"]],
    on="Date",
    suffixes=("_Gold", "_Silver")
)

correlation = comparison["Daily_Return_Gold"].corr(

```

```

    comparison["Daily_Return_Silver"]
)

print(f"Gold-Silver Correlation: {correlation:.2f}")
Gold-Silver Correlation: 0.81

# Create the final Tableau dataset

tableau_data = pd.merge(
    gold_common[
        [
            "Date",
            "Adj Close",
            "Daily_Return",
            "Cumulative_Return",
            "Investment_Value",
            "Drawdown"
        ]
    ],
    silver_common[
        [
            "Date",
            "Adj Close",
            "Daily_Return",
            "Cumulative_Return",
            "Investment_Value",
            "Drawdown"
        ]
    ],
    on="Date",
    suffixes=("_Gold", "_Silver")
)

```

```
tableau_data.head()
```

|   | Price    | Date       | Adj Close_Gold | Daily_Return_Gold |
|---|----------|------------|----------------|-------------------|
| 0 | 0.072122 | 2022-02-07 | 41.630001      | 0.072122          |
| 1 | 0.528849 | 2022-02-08 | 41.820000      | 0.456398          |
| 2 | 0.937508 | 2022-02-09 | 41.990002      | 0.406509          |
| 3 | 1.370192 | 2022-02-10 | 42.169998      | 0.428665          |
| 4 | 1.418270 | 2022-02-11 | 42.189999      | 0.047428          |

| Price | Investment_Value_Gold | Drawdown_Gold | Adj Close_Silver |
|-------|-----------------------|---------------|------------------|
|       |                       |               | \                |

|   |               |     |           |
|---|---------------|-----|-----------|
| 0 | 100072.121623 | 0.0 | 61.770000 |
| 1 | 100528.849108 | 0.0 | 62.279999 |
| 2 | 100937.507737 | 0.0 | 62.770000 |
| 3 | 101370.191624 | 0.0 | 62.799999 |
| 4 | 101418.269650 | 0.0 | 62.360001 |

| Price                     | Daily_Return_Silver | Cumulative_Return_Silver |
|---------------------------|---------------------|--------------------------|
| Investment_Value_Silver \ |                     |                          |
| 0                         | 1.329408            | 1.329408                 |
| 101329.408224             |                     |                          |
| 1                         | 0.825641            | 2.166025                 |
| 102166.025154             |                     |                          |
| 2                         | 0.786772            | 2.969839                 |
| 102969.838975             |                     |                          |
| 3                         | 0.047792            | 3.019050                 |
| 103019.049895             |                     |                          |
| 4                         | -0.700635           | 2.297263                 |
| 102297.262618             |                     |                          |

| Price | Drawdown_Silver |
|-------|-----------------|
| 0     | 0.000000        |
| 1     | 0.000000        |
| 2     | 0.000000        |
| 3     | 0.000000        |
| 4     | -0.700635       |

```
tableau_data["Year"] = tableau_data["Date"].dt.year
```

```
tableau_data.shape
```

```
(1120, 12)
```

```
tableau_data.head()
```

| Price                    | Date       | Adj Close_Gold | Daily_Return_Gold |
|--------------------------|------------|----------------|-------------------|
| Cumulative_Return_Gold \ |            |                |                   |
| 0                        | 2022-02-07 | 41.630001      | 0.072122          |
| 0.072122                 |            |                |                   |
| 1                        | 2022-02-08 | 41.820000      | 0.456398          |
| 0.528849                 |            |                |                   |
| 2                        | 2022-02-09 | 41.990002      | 0.406509          |
| 0.937508                 |            |                |                   |
| 3                        | 2022-02-10 | 42.169998      | 0.428665          |
| 1.370192                 |            |                |                   |
| 4                        | 2022-02-11 | 42.189999      | 0.047428          |
| 1.418270                 |            |                |                   |

| Price | Investment_Value_Gold | Drawdown_Gold | Adj Close_Silver \ |
|-------|-----------------------|---------------|--------------------|
| 0     | 100072.121623         | 0.0           | 61.770000          |
| 1     | 100528.849108         | 0.0           | 62.279999          |
| 2     | 100937.507737         | 0.0           | 62.770000          |

|   |               |     |           |
|---|---------------|-----|-----------|
| 3 | 101370.191624 | 0.0 | 62.799999 |
| 4 | 101418.269650 | 0.0 | 62.360001 |

|                           | Price         | Daily_Return_Silver | Cumulative_Return_Silver |
|---------------------------|---------------|---------------------|--------------------------|
| Investment_Value_Silver \ |               |                     |                          |
| 0                         | 101329.408224 | 1.329408            | 1.329408                 |
| 1                         | 102166.025154 | 0.825641            | 2.166025                 |
| 2                         | 102969.838975 | 0.786772            | 2.969839                 |
| 3                         | 103019.049895 | 0.047792            | 3.019050                 |
| 4                         | 102297.262618 | -0.700635           | 2.297263                 |

|   | Price | Drawdown_Silver | Year |
|---|-------|-----------------|------|
| 0 |       | 0.000000        | 2022 |
| 1 |       | 0.000000        | 2022 |
| 2 |       | 0.000000        | 2022 |
| 3 |       | 0.000000        | 2022 |
| 4 |       | -0.700635       | 2022 |

```
tableau_data.to_csv(
    "Gold_Silver_Tableau_Data.csv",
    index=False
)
```

```
print("Tableau dataset created successfully!")
```

Tableau dataset created successfully!

```
import os
```

```
print(os.path.abspath("Gold_Silver_Tableau_Data.csv"))
```

C:\Users\pc\Desktop\Gold\_Silver\_Tableau\_Data.csv